

Acessing Brazil Central Bank data via API

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Brief guidethrough

This simple library allows the users to access Brazil Central Bank (BACEN) data by accessing its API. In short, there are two R scripts which work jointly: `bacen_url` e `bacen_api`. The script `bacen_url` is responsible for creating the URL where the data is available. The `bacen_api` connects with the API, extracts the info required and converts it to a readable format.

Additionally, the library provides the `bacen_query` function, which combines the functionality of both previous functions into a single interface, simplifying the data retrieval process.

The following sections describe how each function works.

Bacen_URL

```
bacen_url(series_code, start_date, end_date)
```

The mechanism is very simple. The user just have to specify three arguments, which are: `series_code`, `start_date`, `end_date`. These arguments correspond, respectively to the series code, the start and the end date. By specifying these arguments the url's is generated.

In the example bellow, we are considering the Brazilian price index time series (IPCA), whose code is 433, in the 2010/23 interval. Once you called the function, just write the parameters and the job is done.

Attention !

- Have in mind that the date pattern is the brazilian one (*dd/mm/yyyy*);
- Dot not forget to write the dates in string/character format;

```
# ----- #
# --- Example - URL Generation --- #
# ----- #

# --- Bacen_URL Function --- #
source('https://raw.githubusercontent.com/paulo-icaro/Bacen_API/main/bacen_url.R')
↵

# --- Create URL --- #
ipca_br_url = bacen_url(433, '01/01/2010', '31/12/2023') # Generating the URL
ipca_br_url
```

```
[1] "https://api.bcb.gov.br/dados/serie/bcdata.sgs.433/dados?formato=json&dataInicial=01/01/2010&dataFinal=31/12/2023"
```

Bacen_API

```
bacen_api(series_code, start_date, end_date)
```

This function connects with Bacen API by using the *httr* or *httr2* functions. In both cases, the **Bacen_API** function internally checks the HTTP status code. Put in to other words, it verifies whether we get a successful connection (code 200) or a bad connection (code 400 or 404). In case the initial try fails, there are three remaining chances to achieve a valid connection. Some messages are displayed during the download of the data.

The arguments of the function are the *url* and a logical variable named *httr* indicating whether the user wants do use *httr* or *httr2*. The next example shows how to use the function.

```
# ----- #
# --- Example - API Connection --- #
# ----- #

# --- Bacen_API Function --- #
source('https://raw.githubusercontent.com/paulo-icaro/Bacen_API/main/bacen_api.R')

# --- URL --- #
ipca_br_url =
↵ 'https://api.bcb.gov.br/dados/serie/bcdata.sgs.433/dados?formato=json&dataInicial=01/01/2003&dataFinal=31/12/2023'

# --- Acessing API Data --- #
data = bacen_api(url = ipca_br_url, httr = TRUE)
tail(data, n = 10)
```

```
data valor
```

```

243 01/03/2023 0.71
244 01/04/2023 0.61
245 01/05/2023 0.23
246 01/06/2023 -0.08
247 01/07/2023 0.12
248 01/08/2023 0.23
249 01/09/2023 0.26
250 01/10/2023 0.24
251 01/11/2023 0.28
252 01/12/2023 0.56

```

Bacen_Query

```
bacen_query(bacen_series_code, bacen_series_name, start_date, end_date, source_github)
```

This script concatenates both former functions. Besides the already known arguments from **bacen_api** function, the researcher only has to inform two more items in order to proceed with the data extraction. First, one needs to inform the ***series_name*** and also decide whether sourcing the **bacen_url** and **bacen_api** functions from this library.

Let's practice the data extraction:

```

# ----- #
# --- Data Extraction --- #
# ----- #

# --- Bacen API Function --- #
source('https://raw.githubusercontent.com/paulo-icaro/Bacen_API/main/bacen_query.R')

# --- Previous Info --- #
cod_bacen_series = c('25390', '4189', '3697', '433')
name_bacen_series = c('ibcrce', 'selic_acum_anual', 'tx_cambio_mp', 'inflação_ipca')
start_date = '01/01/2015'
end_date = '31/12/2025'

# --- Extraction --- #
bacen_dataset = bacen_query(cod_bacen_series, name_bacen_series, start_date,
↪ end_date)
tail(bacen_dataset)

```

```

      data ibcrce selic_acum_anual tx_cambio_mp inflação_ipca
127 01/07/2025 114.73             14.90         5.5279         0.26
128 01/08/2025 115.55             14.90         5.4463        -0.11
129 01/09/2025 114.29             14.90         5.3668         0.48

```

130	01/10/2025	112.62	14.90	5.3849	0.09
131	01/11/2025	107.25	14.90	5.3403	0.18
132	01/12/2025	106.77	14.90	5.4525	0.33